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Scaling up by looking closer: extending the range of moss-Nostoc symbioses deeper into central Europe and to new moss hosts in the forest canopy --Manuscript Draft--

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Corresponding Author:	Daniel Thomas, Ph.D. University of Bayreuth: Universitat Bayreuth Bayreuth, Bayern GERMANY	
Order of Authors:	Martin Feulner, PhD Ulla Spieleder Frederic J. Pascual Dose Jona Motzkus Daniel Thomas, Ph.D.	
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Bayreuth Center of Ecology
and Environmental Research

bayceer



Dr. Daniel Thomas

Akademischer rat a.Z.
Lehrstuhl f. Ökologische
Mikrobiologie
Universität Bayreuth

Telefon +49(0)921/55-5620

e-mail: daniel.thomas@uni-bayreuth.de

<https://www.bayceer.uni-bayreuth.de/bayceer/>

Dr. Hans-Frisch-Straße 1-3
95448 Bayreuth,
Germany

To the Editorial team of Oecologia,

Please find attached our manuscript, “Scaling up by looking closer: extending the range of moss-*Nostoc* symbioses deeper into central Europe and to new moss hosts in the forest canopy.”

In this project we searched for – and found – the well-known feather-moss/*Nostoc* symbioses in our region of Germany. As far as we are aware, we are the first to provide detailed observations of this symbiosis in our region and in Germany, complete with microscopic evidence and genetically identified cultures of the microorganisms involved. The likely presence of *Nostoc* in forest mosses has been incidentally noted in one environmental metabarcode study (see our introduction), but was not further confirmed.

Regardless of the possible geographical extension of this symbioses, we also make several other interesting observations. We used Nanopore sequencing to observe the unexpected presence of multiple possible *nifH* genes within our *Nostoc* cultures, which causes us to call for the re-interpretation of the results of important previous ecological studies on moss-*Nostoc* *nifH* studies by other groups. We contextualize the feather-moss/*Nostoc* symbioses into the increasing awareness that all mosses generally host a rich community of cyanobacteria. We present PCR-based evidence for the possible presence of cyanobacterial nitrogenases throughout the aerial tissues of a feather mosses, despite lack of visible colonization by *Nostoc*, and use epifluorescent microscopic to provide a possible explanation for this: possible cyanobacterial endophytes in the “stems” (caulids) of our mosses.

We hope Oecologia finds this manuscript of sufficient interest for its readers. We look forward to your response.

Sincerely,

Daniel Thomas
19 January 2026

Scaling up by looking closer: extending the range of moss-Nostoc symbioses deeper into central Europe and to new moss hosts in the forest canopy

Feulner Martin, Ulla Spieleder, Frederic J. Pascual Dose, Jona Motzkus, Daniel C. Thomas

Abstract

Since their re-discovery, Moss-cyanobacterial symbioses are of increasing interest, especially *Nostoc*-feather-moss symbioses. Efforts to document *Nostoc*-feather-moss symbioses have historically been centered in Scandinavia and the North American Boreal forest zone. Herein we report the results of several opportunistic surveys investigating the occurrence of *Nostoc*-moss symbioses in the region of Franconia (Germany). Using culture- and microscopy-based methods, we confirm the occurrence of such symbioses in our region. Two new moss host species of the epiphytic family Orthotrichaceae, *Orthotrichum* and *Zygodon*, are also documented, further extending the range of known moss hosts for *Nostoc*-like organisms outside the traditional feather moss (Hypnales) and *Sphagnum* clades. Full length 16S sequence alignments placed the moss-associated *Nostoc*-like organisms throughout our *Nostocaceae* phylogeny, with no obvious single symbiotic clade. We note the presence of multiple alternate *nifH* genes within the genomes of single *Nostoc*-like organisms, possibly enabling diazotrophy under variable ecological conditions. We also present evidence for possible endophytic colonization by cyanobacteria of stems of mosses.

Keywords

Nostocaceae, Hypnales, Orthotrichaceae, endophyte, cyanobacteria

22 Introduction

23 Close associations between *Nostoc* cyanobacteria and mosses have been noted by naturalists
24 for over a century. Among the first described species of *Nostoc* (sensu lato, *Nostoc* Vaucher ex
25 Bornet and Flahault 1886) were specimens collected from mosses (*Nostoc muscorum* Agardh
26 ex Bornet et Flahault 1888, syn. *Desmonostoc muscorum* (Hrouzek et al., 2013). Researchers
27 throughout the twentieth century continued to note frequent associations of *Nostoc*-like
28 organisms and mosses (division Bryophyta) (Meeks, 1990). The potential importance of these
29 associations at larger scale in temperate and arctic terrestrial ecosystems was predicted and
30 quantified in multiple early studies (Dodds et al., 1995). While all mosses are known to host
31 cyanobacterial communities (Groß et al., 2024; Rousk, 2022), especially *Sphagnum* mosses
32 (Rousk et al., 2015), interest in bryophyte-*Nostoc* associations grew rapidly following the
33 documentation of a widespread symbiosis between a *Nostoc* sp. and a feather moss
34 *Pleurozium schreberi* in boreal forests (DeLuca et al., 2002). The super-abundance of
35 *Pleurozium* and other feather mosses (order Hypnales) in the large northern forest zones
36 implied immense importance for global-scale N and C budgets. Since these findings, research
37 on feather-moss/*Nostoc* symbioses has accelerated and studies have employed increasingly
38 sophisticated means of detection and quantification of nitrogen-fixation and nutrient transfer
39 (Alvarenga et al., 2024; Arróniz-Crespo et al., 2022; Bay et al., 2013; Stuart et al., 2020).

40 Moss-cyanobacterial mutualisms appear to strongly contribute to the primary production of
41 boreal forests and therefore contributes to ecosystem services (timber production) and
42 sustainability (reduced fertilizer inputs). Through associations with nitrogen-fixing
43 microorganisms, mosses can provide large inputs of nitrogen for ecosystems. In boreal forests,
44 up to 50% of the nitrogen input may come from cyanobacteria associated with mosses (Rousk &
45 Michelsen, 2017). In peatland cyanobacteria can contribute more than 35% of the nitrogen
46 required by mosses like *Sphagnum*, helping maintain the productivity and stability of these

47 carbon-rich environments (Berg et al., 2013). In harsher climates, such as tundra regions
48 (reviewed by Lindow et al. (2013)) or tree-canopy microclimates (Lindo & Whiteley, 2011),
49 cyanobacteria associated with mosses can account for the vast majority of the nitrogen-fixation
50 necessary for observed rates of net primary production. Variation in the community of nitrogen-
51 fixing bacteria -- and especially cyanobacteria -- appears to be explained by both geography
52 (Groß et al., 2024) and the host specificity (Ininbergs et al., 2011).

53 Following classical, broad definitions of symbiosis as any long-term, physically-proximal biotic
54 interaction (Tipton et al. 2019), symbioses between feather mosses and Nostoc-like organisms
55 are generally considered to be “loose”, mutually-facultative or commensal exosymbioses.
56 Nostoc-like organisms appear to prefer to epiphytically inhabit the spaces in protected leaflet-
57 axils along moss caulids (“stems”), or in incurved regions of leaflet-tissue and other protected
58 areas (DeLuca et al., 2002; Solheim et al., 2004), suggesting at least a structural benefit for
59 Nostoc-like organisms when associating with moss hosts. Indeed, the main advantage for the
60 cyanobacterium may be a buffered environment on the moss, as the hygroscopic nature of
61 mosses may create sheltered microenvironments even during times of high radiative exposure
62 (Maeng et al., 2025) or drought (Li et al., 2024) for their resident microbiome.

63 Benefits from the moss-*Nostoc* symbioses for the moss host are presumed, as chemo-attraction
64 and induction of mobile hormogonia phase by feather-mosses in Nostoc symbionts have been
65 clearly documented (Bay et al., 2013), at an energetic cost to the host moss (Sprent & Meeks,
66 2013). Nutrient exchanges are hypothesized, involving a transfer of fixed organic nitrogen in the
67 form of fixed, organic nitrogen from heterocystous Nostoc colonies to the host plant (Arróniz-
68 Crespo et al., 2022; Bay et al., 2013), perhaps in exchange for organic carbon- and sulfur-
69 containing materials from the moss host, though the latter has only recently been directly
70 reported (Stuart et al., 2020). Symbioses vary seasonally and daily (Warshan et al., 2016) in
71 their levels of nitrogen-fixing activity and microbial abundance, due to climatic constraints on

72 microbial metabolism (Gentili et al., 2005) and N-availability at both host (Renaudin et al., 2022)
73 and ecosystem scales (Cusack et al., 2009; Zheng et al., 2019).



74 **Figure 1:** *Hylocomium splendens* with colonies of *Nostoc* ((c) J. Motzkus 2024)

75 Traditionally, *Nostoc*-moss symbioses have been studied most intensively in Scandinavian and
 76 North American regions (Alvarenga & Rousk, 2022), but they have been observed in many
 77 extraordinary sites throughout the world, including retreating glacial fields of Tierra del Fuego
 78 (Chile) (Arróniz-Crespo et al., 2014), Andean tropical montane cloud forest (Peru) (Permin et al.,
 79 2022) and Mount Fuji (Japan) (Kubota et al., 2023). Arboreal mosses are increasingly of
 80 interest, and have been observed to be hotspots for moss-cyanobacterial symbioses (Lindo &

81 Whiteley, 2011). Despite its proximity to Scandinavia and despite many shared moss species
82 and sites of similar environmental conditions, direct observations of the symbiosis of feather
83 mosses and *Nostoc* (or *Nostoc*-like organisms) are rare from the Central European region. To
84 our knowledge the *Nostoc* symbiosis has been observed in German mosses only once, as one
85 member out of many nitrogen-fixing bacteria associated with log-dwelling moss species, using
86 *nifH* metabarcoding (Groß et al., 2024). Traditional “model” species of feather-moss involved in
87 *Nostoc* symbioses like *Hylocomium* (Fig. 1) or *Pleurozium* have to date received very little
88 attention in central Europe, despite the prevalence of these moss species in montane forests of
89 central Europe. Particularly likely to host moss-*Nostoc* symbioses are spruce- or pine-
90 dominated forests, as *Pleurozium schreberi* or *Hylocomium splendens* are among the dominant
91 floor mosses in these forests (Ellenberg, 1988).

92 Here we sought to confirm the presence of the moss-*Nostoc* symbioses in the region of
93 Franconia, Germany, and to further explore the known host range for the symbiosis. We
94 examined mosses from the Fichtel mountains, the Frankenjura, and adjacent areas, concurrent
95 with on-going moss diversity inventories and monitoring efforts in the region. Special attention
96 was given to arboreal epiphytic mosses, especially in the family Orthotrichaceae. despite never
97 having been documented as hosts for the symbiosis. Members of Orthotrichaceae are best
98 known as epiphytic mosses from tree bark (Fialová et al., 2023), in our region especially on
99 broad-leaved trees with many species. We hypothesized them as likely candidates to host the
100 symbiosis, given the presumed low nitrogen availability and dynamic microclimate of the canopy
101 setting. We utilized a microscopy- and culture-based approach for detection and description of
102 *Nostoc*-like organisms, which we hoped would enable precise imaging and taxonomic
103 placement of symbionts, using full-length 16S and partial *nifH* PCR amplicon sequence data.

104 Methods

105 *Field*

106 Mosses from multiple sites through the region of upper and middle Franconia (Germany) were
107 opportunistically sampled for *Nostoc*-like organisms, as part of ongoing moss diversity
108 inventories in the region. They were determined using standard literature (Frahm, Frei , Lüth
109 2011) Moss specimens were collected when visual inspection with a 10x lupe indicated possible
110 cyanobacterial colonies, in the form of blue discoloration. Moss samples were collected from
111 multiple different locations in the Fichtel Mountains, the Frankenjura, and other sites at
112 elevations between 350 and 800 m above sea level. Sample collections were carried out as part
113 of other ecological surveys which took place during the years 2023 - 2025, see Table 1 for
114 details. Each moss sample consisted of at least one healthy gametophyte individual (all
115 continuous tissue from the rhizoids to the shoot). Samples were stored in plastic bags and kept
116 in the dark at 4 °C before further use.

117 Table 1. Sampled locations.

Locality name	Latitude/ longitude (°)	Moss species	substrate	Elev. (m)	Sample date
Waldstein, Fichtelgebirge	50,14233 11,86628	Hylocomium splendens	soil	774	May-2023 May-2024
University of Bayreuth	49,92438 11,58454	H. splendens	soil	354	May-2023
Northern Fraco- nian Jura, bei Neustädtlein	49,96020 11,43589	Orthotrichum lyellii	Bark (Fraxinus excelsior)	480	Nov-2024
Pfaffenberg bei Mettendorf	49,02075 11,35922	Zygodon rupestris	Bark (Sorbus sp.)	470	Nov-2024
Schwarzachk- lamm bei Gsteinach	49,352118 11,232484	Homalia trichomanoides	Bark (Fagus sylvatica)	345	May-2025
Pfaffenberg bei Mettendorf	49,02075 11,35922	Pseudoamblystegium subtile	Bark (Sorbus sp.)	470	Nov-2024

118 Microbiology:

119 Overview:

120 All moss samples were screened microscopically for colonization by *Nostoc*-like species, first
121 with classical light microscopy, followed by additional wide-field epifluorescence microscopy in
122 specimens where colonies were observed. Following microscopy, culture of all observed
123 spherical *Nostoc*-like colonies was attempted, and amplification of *NifH* and 16S rRNA marker
124 genes, using colony PCRs. Whole community DNA (moss and microbiome) was extracted for
125 some samples for PCR-based detection of *nifH* in moss tissues. In some cases, surface-
126 sterilization was performed to detect the possibility of endophytic colonization by cyanobacteria
127 of moss tissues.

128 Microscopy

129 Preliminary surveys for the presence of *Nostoc* were conducted with standard light microscopy,
130 first using binocular stereoscopes (Wild HeerBrug M-series) and later using Zeiss compound
131 microscopes. For comparison, some specimens which were surveyed for *Nostoc* colonies using
132 light microscope and that were found not to host visible colonies were then subjected to DNA
133 purification and to whole-community PCR amplification of *nifH* genes in PCR tests. Sectioning
134 was done using sections of one centimeter from the top (o), middle (m) and bottom (u) of
135 caulids (Fig. 1), for comparison of relative abundances of *nifH* abundances among different
136 positions. Samples with possible *Nostoc* colonies were then subjected to further investigation
137 with epifluorescence microscopy.

138 Moss tissues and bacterial cultures were also observed under an epifluorescence microscope
139 (Revolve 2, ECHO, San Diego, USA), using Revolve Software 4.1.0, app version v7.1.0. With
140 epifluorescence microscopy, cyanobacteria can be easily more discerned from eukaryotic
141 microalgae and plant cells, as cyanobacteria present a distinct autofluorescent “fingerprint”
142 characterized by phycobilisome-mediated fluorescence (Grigoryeva, 2020; Wojtasiewicz &
143 Stoń-Egiert, 2016). As such, cyanobacterial cells can be identified using the standard “Cy5” filter
144 setup (CHROMA 49009, ex: 610-650 nm, em: 660-735 nm), which detects C- phycocyanins
145 (phycocyanins found only in cyanobacteria) with an excitation peak at around 625 nm (em: ≥650
146 nm) (Grigoryeva, 2020; Poniedziałek et al., 2017). In contrast, chlorophyll a fluorescence shows
147 an excitation peak at 458 nm and ~680 nm (em: ~680 nm). As such, algal and plant cells —
148 which contain significantly higher concentrations of chl a and no C-phycocyanin; (Wojtasiewicz
149 & Stoń-Egiert, 2016) — can be detected with the “FITC/Cy2” filter set (CHROMA 49002, ex:
150 450-490 nm, em: 500-550 nm), which permits some detection of emissions from Chl a and
151 select other embryophyta plant cell components, especially in cell walls (Donaldson, 2020;
152 Tobimatsu et al., 2013). Phycoerythrin, which is present in varying concentrations in different

141 cyanobacterial species, can also be detected using the “TXRED” filter (CHROMA 49008, ex:
142 540-580, em: 590-665 nm).

143 Surface sterilization

144 To test the hypothesis that cyanobacteria inhabit endophytic compartments of the host mosses,
145 some *Hylocomium* moss samples were surface sterilized. For surface sterilization, Alvarenga et
146 al. (2024), was followed, with some modifications: 2-3 grams (about 5-10 gametophytes) of
147 moss were manually cleaned of debris, placed into plastic 50mL centrifuge tubes, and
148 submerged in 40mL of 0.5% Tween20 solution. Samples were ultrasonicated for a minimum of
149 2 minutes with a VWR ultrasonic cleaner at power 5. Liquid was removed by decanting, and
150 replaced with 40 mL sterile de-ionized, distilled H₂O, vortexed for 10 seconds, and discarded by
151 decanting. Samples were then subjected twice to submersion in 1% NaOCl solution, and
152 agitation by vortex mixer for 10 seconds at high speed. After each NaOCl application, samples
153 were rinsed and agitated in 40 mL sterile de-ionized, distilled H₂O. Surface sterilized samples
154 were then again inspected for visible cyanobacterial colonization.

155 Culture of *Nostoc*-like organisms

156 The abundant chloroplast DNA present in host moss tissue complicates molecular laboratory
157 and bioinformatic pipelines for characterizing cyanobacteria (Fitzpatrick et al., 2018; Song & Xie,
158 2020), so we attempted a cultivation-based pipeline for describing our observed *Nostoc*-like
159 organisms during microscopic inspection. “*Nostoc*-like organisms” were defined as any
160 filamentous bacteria that were observed in sheathed syncoenobia, especially when trichomes
161 displayed obvious intercalary heterocytes. Media for all cultivation was nitrogen-free BG-110
162 media (Rippka et al., 1979). Entire stems (0.5-2 cm length) of host moss were submerged into
163 liquid media for 2-8 weeks in 15 mL Falcon centrifuge tubes at room temperatures with natural

(window) lighting, followed by removal of moss tissue upon observation of phototrophic growth, to reduce further contamination. Artificial augmentation of both temperature and light levels were attempted but either did not produce additional growth or resulted in colony death. Most phototrophic growth observed in these samples were due to eukaryotic algae, but in some cases growth of Nostoc-like colonies was observed. These colonies of Nostoc or Nostoc-like organisms were then further separated into separate liquid media aliquots through trapping with pipette tips, retaining as little host tissue and Eukaryotic algae as possible in the process. After 1-2 months, these colonies produced additional Nostoc colonies of the classic, spherical morphology, which were then individually captured using micropipettes set to take up 2 microliter volumes. Colonies were then transferred to new 1.5 mL centrifuge tubes, where they were subjected to a sterile water wash and subsequent submersion in fresh BG-110 media in 15 mL centrifuge tubes. These isolations were allowed to grow an additional 1-2 months, then were subject to microscopy as detailed above, and colony PCR as detailed below. As of time of writing, Nostoc-like organisms were successfully isolated and cultivated from 4 separate moss host species, for further DNA extraction and PCR.

DNA extraction

Whole-community (host and microbial) DNA was extracted using the Quick-Start Protocol DNeasy®Plant Pro Kit from QIAGEN®, following manufacturer instructions. In step 1, twice the amount of CD1 was initially added, as the specified amount of added liquid was often not sufficient to rehydrate moss tissue.

For use as additional PCR controls for whole-community *nifH*, DNA was also extracted from (1) liquid culture of *Anabaena variabilis* (DSM 107003), (2) leaf and stems of surface-sterilized moss *Brachythecium* sp. collected from a roadside urban grass-dominated landscape in Bayreuth, Germany, that was presumed to be abundant with nitrates and therefore low in cyanobacterial nitrogen fixation activity, and (3) surface-sterilized leaf tissue from *Solidago* sp.,

189 a common vascular plant in the region. After extraction, the DNA was checked for quality and
190 quantity using the NanoDrop One (Thermo Scientific) and Qubit 4 assay (Thermo Fisher
191 Scientific) and frozen at -20°C to preserve it for later testing.

192 PCR

193 For amplification of *nifH* gene, the CNF (5'-CGTAGGTTGCGACCCTAAGGCTGA-3') and CNR
194 (5'-GCATACATCGCCATCATTTTACC-3') primers (Olson et al., 1998) were used as per Díez et
195 al. (2007). These primers are expected to amplify sequences of 359 bp length when used on
196 cyanobacterial DNA. *NifH* genes from DNA of whole-community moss+microbiome were
197 amplified from samples of *H. splendens* collected from the University of Bayreuth Ecological
198 Botanical Garden. Reactions were conducted in 25 µL volumes: 12.5 µL NEB Q5® High-Fidelity
199 2X Master Mix, 0.3 µL BSA (20 ng/µL), 1.2 µL forward and reverse primers each (10 µM), 7.3
200 µL PCR-grade H₂O and 2.5 µL of template DNA. PCR conditions were as follows: initial
201 denaturation at 98°C for 2 min, followed by 32 cycles of 98°C denaturing for 15 s, 50°C
202 annealing for 30 s and 72°C extension for 30 s, and a final extension of 72°C for 7 min.

203 As isolation and growth of *Nostoc*-like colonies was extremely slow and yielded very little
204 biomass for DNA extraction, direct colony PCR of individual *Nostoc* colonies was necessary to
205 target and amplify full length 16S rRNA and *nifH* genes. Individual *Nostoc*-like spherical colonies
206 were trapped via suction onto 20 µL pipette tips, washed in sterile water, and placed directly into
207 50 µL reaction volumes. PCR reagents were mixed as follows, per 50 µL reaction: 10 µL NEB
208 Q5® 5× reaction buffer, 10 µL NEB Q5® High GC enhancer, 0.5 µL NEB Q5® HiFi polymerase,
209 1.0 µL mixed dNTPs (10 mM), 0.6 µL BSA (20 ng/ µL), 1.2 µL forward and reverse primers
210 each (10 µM), 4 µL of template DNA, and PCR-grade H₂O to 50 µL total volume. PCR
211 conditions were as follows: initial denaturation at 98°C for 5 min, followed by 30 cycles of 98°C
212 denaturing for 15 s, 50°C annealing for 30 s and 72°C extension for 30 s, and a final extension
213 of 72°C for 7 min.

214 For 16S rRNA-targeted PCR on *Nostoc*-like colonies, protocol followed Cuscó et al., (2019),
215 with modifications for the colony-PCR. Primers 16S-27F (5'-agrgtttgatyhtggctcag-3') (Zeng et al.
216 2013) and 16S-1492R (5'-taccttgtaggactt-3') (Klindworth et al., 2013) were used, targeting the
217 majority of the 16s rRNA gene. PCR reagents were mixed as follows, per 50 µL reaction: 10 µL
218 NEB Q5® 5× reaction buffer, 10 µL NEB Q5® High GC enhancer, 0.5 µL NEB Q5® HiFi
219 polymerase, 1.0 µL mixed dNTPs (10 mM), 0.4 µL BSA (20 ng/µL), 2.5 µL forward and reverse
220 primers each (10 µM), 4 µL of template DNA, and PCR-grade H₂O to 50 µL total volume. PCR
221 conditions were as follows: denaturation at 98°C for 4 min amplification, followed by 30 cycles of
222 98°C for 15 s, 51°C annealing for 30 s, and 72°C extension for 1 min, and a final extension of
223 72°C for 7 min. PCR products were cleaned using a Nucleospin Gel and PCR cleanup column
224 kit (Macherey-Nagel, www.mn-net.com, article-number: 740609.50), with eluted to a 14 µL
225 volume, with final DNA concentrations between 3.8 ng/µL - 40.2 ng/µL.

226 Sequencing of *nifH* and 16S rRNA genes

227 Colony-PCR products from *Nostoc*-like cultures were then prepped according to manufacturer
228 instructions for Nanopore-platform sequencing using either (1) the Oxford Nanopore
229 Technologies (ONT) ligation sequencing kit V14 (SQK-LSK114) or (2) ONT Rapid sequencing
230 V14 -Amplicon sequencing (SQK-RBK114.24), with a R10.4.1 (FLO-MIN114) generation flow
231 cell. SQK-LSK114 kit was used in the case of 16S rRNA gene of *Hylocomium*-associated
232 *Nostoc*, and the remaining 6 colony-PCR products were individually barcoded, multiplexed and
233 sequenced jointly using a SQK-RBK114.24 kit. In the first sequencing run, with only
234 *Hylocomium*-associated *Nostoc*, sequencing ran until a depth of ~100,000 reads, with
235 simultaneous fast-model basecalling using Guppy (MinKNOW v23.07.12, Bream v7.7.6, Guppy
236 v7.1.4, MinKNOW Core v5.7.5). In the second, multiplexed sequencing run sequencing was run
237 until total depth of ~38,000 reads, without basecalling, which was conducted separately with

238 Dorado basecalling software (MinKNOW v25.05.12, Bream v8.5.4, Dorado v7.9.8, MinKNOW
239 Core v6.5.13). In both cases default minimum Q-scores of 8 were used to define passing reads.

240 Bioinformatics

241 Processing raw reads to consensus sequences
242 Demultiplexing and trimming of adapters and barcodes of Nanopore reads from colony-PCRs of
243 Nostoc-like cultures was done using the Dorado basecalling software. Resulting reads were then
244 re-oriented 5'-3' where necessary, using VSEARCH (Rognes et al., 2016) v2.30.0, against the
245 Silva 138.2 (Quast et al., 2013) Genus-level training dataset, formatted for DADA2 software
246 ("silva_nr99_v138.2_toGenus_trainset.fa", available at
247 www.arb-silva.de/current-release/DADA2/1.36.0/SSU). Re-orientation of nifH reads was
248 conducted against the custom nifH database of nifH sequences from Kubota et al. (2023).
249 Resulting reads from nanopore sequencing were typically numerous and very noisy, aligning
250 poorly due to numerous erroneous insertions, paralogs, and co-cultured microorganisms. In
251 some cases, BLAST(Basic Local Alignment Search Tool)-family algorithms (Altschul et al.,
252 1990) have been found to perform effectively for aligning reads with common error patterns of
253 long-read metabarcoding data (Tedersoo et al., 2021, 2022). Therefore, as a first filtering step
254 before full alignments, we used the BLAST-suite tools, specifically `blastn` and `blastx` programs to
255 find high quality matches to Nostoc 16S sequences from the Silva 138.2 database (see link
256 above), and to a custom database of nifH amino acid sequences. Following Hu, Wang, et al.
257 (2024), our custom database of nifH sequences was a concatenation of NifH sequences from
258 (1) 963 culture-derived genomes (Koirala and Brözel, 2021), the Greening lab NifH database
259 (<https://doi.org/10.26180/c.5230745>) and the Nathoo database (Suquilanda-Pesántez et al.,
260 2022). Following this filtering step, reads were aligned de novo using MUSCLE 5.1 with default
261 settings (Edgar, 2022). Vertical gaps were removed using CAlign (Tumescheit et al., 2022),

262 using a insertion minimum percentage of 70%, of all gap sizes (insertion minimum size > 0,
263 insertion minimum flank size > 0).

264 Both 16S (Sun et al., 2013) and nifH (Thiel, 1993; Thiel et al., 1995) can be present in multiple
265 unique copies within a single organism. To detect this possible paralogous, intra-genomic
266 variation, trees of full alignments of amplicon sequences were constructed using FastTree 2.2
267 with default settings (Price et al., 2010), and visualized with ARB 7.0 (Ludwig et al., 2004).
268 Unrooted trees were searched for obviously separate clades (example shown in Supp. Fig. 1). If
269 present, these reads from these subtrees were again aligned individually using MUSCLE, and
270 manually curated for removal of poorly aligned sequences and ends. If obviously separate
271 clades were not present in the phylogenetic tree, the full alignment of 16s or nifH was assumed
272 to represent a single representative gene sequence for that organism. In either case, consensus
273 sequences were generated using Hidden Markov models (HMM) from each (sub)alignment,
274 using the `hmmbuild` and `hmmemit` commands, from the HMMER v3.3 package (Eddy, 2011),
275 using simple majority-rule decision for consensus-sequence bp calls. Since nifH Nanopore
276 sequences were not preliminarily taxonomically filtered to Nostoc, unlike 16S (see above),
277 resulting nifH consensus nucleotide sequences were at this point searched against the full NCBI
278 Core nucleotide BLAST database (blast.ncbi.nlm.nih.gov), for highly similar matches using the
279 online blastn graphical user interface, to confirm likely Nostocaceae origin.

280 NifH phylogeny

281 Our nifH phylogenetic analysis was intended to follow - and build upon - Ininbergs et al. (2011)
282 and Kubota et al. (2023). All pertinent nifH sequences from both studies were provided by
283 authors (Masayuki Kubota, pers. comm.), combined with nifH consensus sequences from our
284 cultures. Alignment was conducted using Muscle v5.1 with default settings, and tree was
285 constructed using FastTree2 using default settings.

286 16S Nostocaceae phylogeny

287 For taxonomic identification of full-length 16S consensus sequences from cultured *Nostoc*-like
 288 organisms, a de novo phylogenetic tree of full-length 16S sequences for Nostocaceae was
 289 constructed. The full Cyanoseq (Lefler et al., 2023) v1.2 phylogenetic tree
 290 (<https://zenodo.org/records/7864137>) was visualized in ARB v7.0, and sequences of all
 291 members of Nostocaceae were selected, including sequences from closely allied clades
 292 Tolypotrichaceae, Aphanizomenaceae, and Nodulariaceae (Strunecký et al., 2023).
 293 Additionally, 708 sequences of full or near-full length 16S from NCBI genbank that were
 294 identified as high-quality *Nostoc* sequences by Cyanoseq v1.3 were provided by the authors for
 295 use in our 16S tree (Forrest Lefler, pers. comm; full cyanobacterial dataset available at
 296 <https://zenodo.org/records/13910424>), along with several of 16S reference sequences of
 297 interest: *Nostoc azollae* (NC_014248.1) (Ran et al., 2010), *Nostoc calcicola* SAG 1453-1
 298 (KM019926) and *Nostoc insulare* SAG 54.79 (KM019927). Full 16S sequence of *N. azollae* was
 299 not available as a stand-alone NCBI nucleotide accession, so 16S copies were extracted from
 300 reference genome with barrnap (<https://github.com/tseemann/barrnap>). *Scytonema*
 301 *pachmarhiense* (MH260366) was added as an outgroup for rooting of phylogenetic trees. In
 302 total, 856 sequences were included in our Nostocaceae tree. Reads were aligned using SSU-
 303 align v0.1.1 (Nawrocki, 2009), using default Bacterial models. Tree was constructed using
 304 FastTree2. Full alignment FASTA and tree Newick files are available in supplementary
 305 materials. Tree topology was visualized with ARB 7.0, and final cosmetic additions to tree labels
 306 and highlighting of taxa of interest were done using Inkscape v1.3.2 for GNU linux.

307 Results

308 Microscopy

309 *Nostoc*-like organisms were observed in 5 moss species, from 4 sites (Figures 2-6):

310 *Orthotrichum lyelli* from Northern Fraconianjura bei Neustädtlein, *Hylocomium splendens* from
311 Waldstein, *Zygodon rupestris* and *Pseudoamblystegium subtilis* from Pfaffenberg bei
312 Mettendorf, and *Homalia trichomanoides* from Schwarzachklamm bei Gsteinach. Within each
313 moss, position of *Nostoc*-like colonies were variable but were most common in the leaflet axils
314 of the moss gametophytes. Most notably in *Orthotrichum lyelli*, the position of *Nostoc*-like
315 colonies were more variable, occurring in the leaf axils but also often on leaf laminas.



Figure 2. Colonies of *Nostoc*-like organism observed from *Orthotrichum lyelli* at 400× magnification.

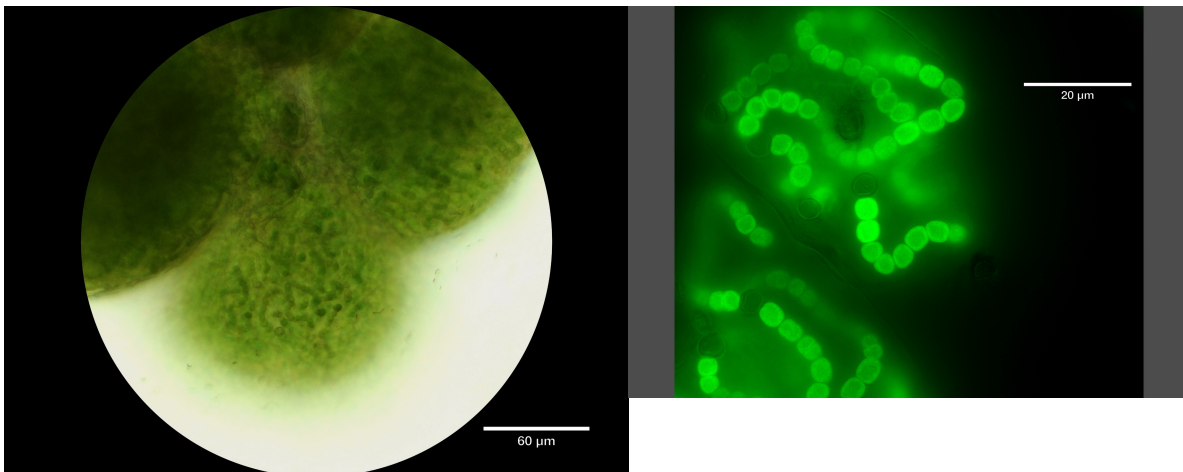


Figure 3. Colonies of *Nostoc*-like organism after isolation from *Hylocomium splendens*. (a) light microscopy, at 400× magnification and (b) 1000× magnification using epifluorescence with Cy5 filter setup (see methods), showing trichomes and heterocysts.

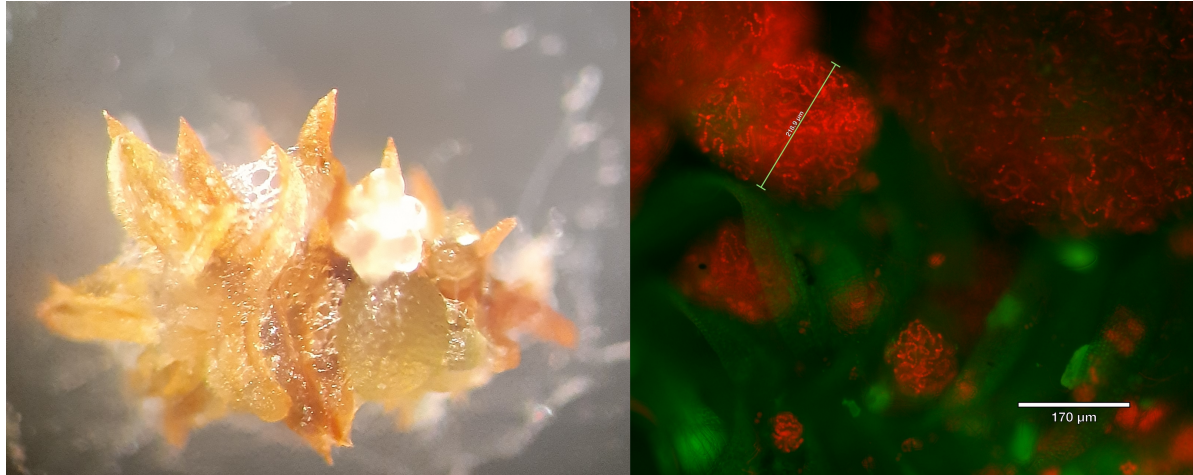


Figure 4. Colonies of *Nostoc*-like organism observed from *Zygodon rupestris* (a) at 50× magnification with binocular dissecting microscope show heavy colonization of moss caulid. (b) at 400×, same colonies using using epifluorescence with Cy5 filter setup for detection of cyanobacteria, overlaid onto FITC setup for background moss tissue (see methods).

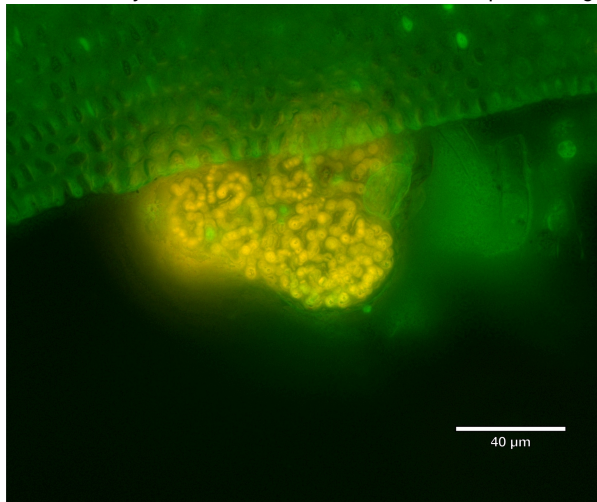


Figure 5. Colonies of *Nostoc*-like organism observed from *Homalia trichomanoides* at 400× using epifluorescence with TXRED filter setup for detection of cyanobacteria, overlaid onto FITC setup for background moss tissue (see methods).

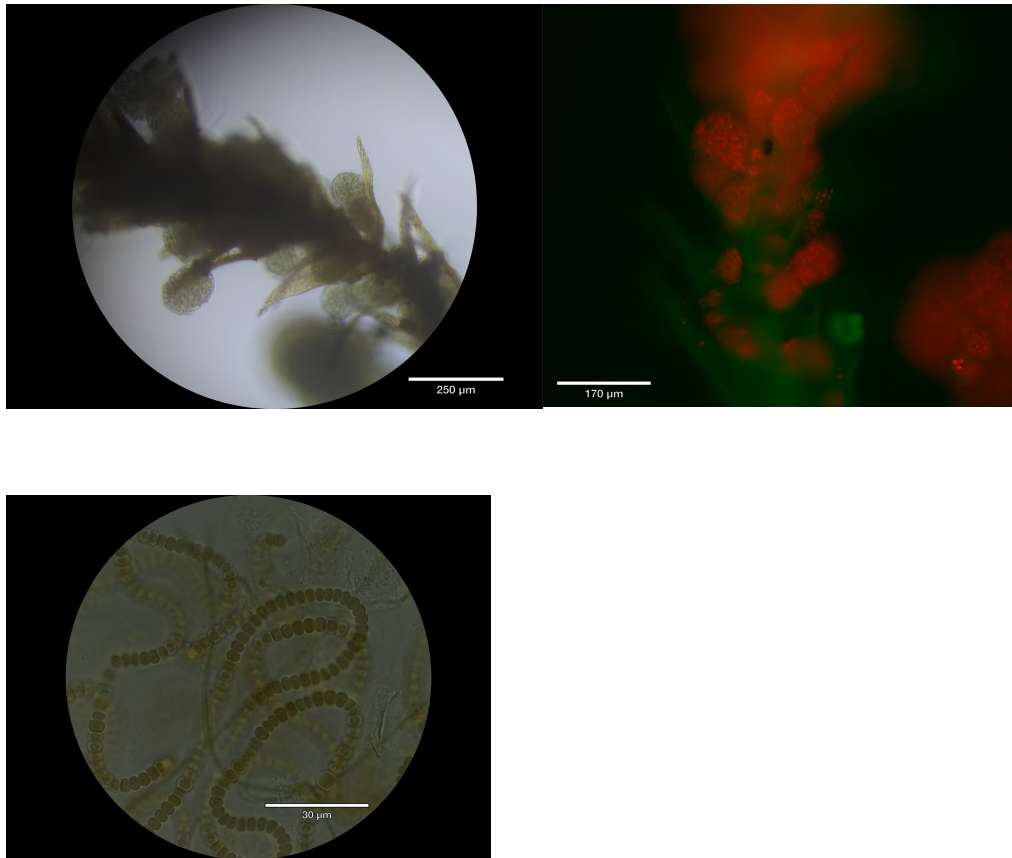


Figure 6. Colonies of *Nostoc*-like organism observed from *Pseudoamblystegium subtile*. (a,b) abundant colonization by nostoc colonies on caulid at 100× magnification with light and using epifluorescence with Cy5 filter setup for detection of cyanobacteria, overlaid onto FITC setup for background moss tissue (see methods). (c) 1000× magnification of same organism, showing trichomes and heterocysts.

316 Surface sterilization

317 Surface sterilization of *Hylocomium splendens* fronds greatly reduced the amount of visible
 318 colonization by cyanobacteria and other phototrophs (Fig. 7). Generally, surface sterilized leaf
 319 surfaces were cleared of visible colonies, though occasionally some microbial cells were
 320 observed (Fig. 7c). *H. splendens* stems still appeared to consistently harbor extensive
 321 significant microbial biomass following surface sterilization, possibly cyanobacteria because of
 322 their fluorescence patterns (Fig. 7d).

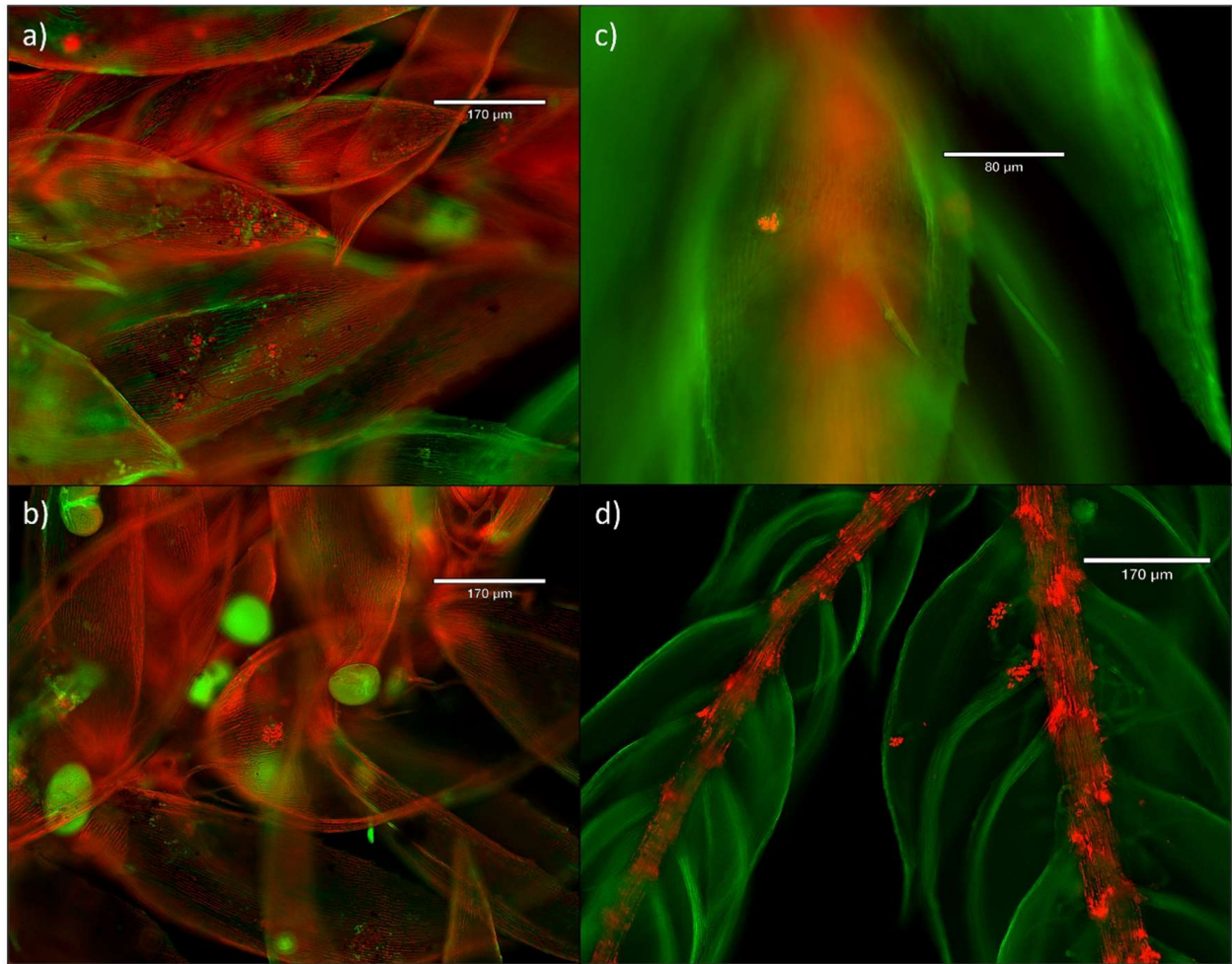


Figure 7. Microbiota of leaves and stems of *Hylocomium splendens*, using epifluorescence with Cy5 filter setup for detection of cyanobacteria, overlaid onto FITC setup for background moss tissue (see methods). (a-b) fluorescence of cyanobacterial and eukaryotic algae on untreated moss leaves. (c,d) after surface sterilization, eukaryotic-algal and cyanobacterial cells on leaves are mostly no longer visible, but colonies on stems remain visible, possibly indicating endophytic lifestyle within stems.

323 Culture of *Nostoc*-like organisms

324 Out of 5 observed *Nostoc*-like organisms, 4 strains were successfully isolated. Successful
 325 *Nostoc*-like isolates were from: *Hylocomium splendens* (Fig. 3), *Homalia trichomanoides* (Fig.
 326 5), *Pseudoamblystegium subtile* (Fig. 6), and *Zygodon* sp. (Fig. 4). The *Nostoc*-like organism
 327 from *Orthotrichum* moss (Fig. 2) failed to show any growth and typically became chlorotic,
 328 despite abundant growth on the collected moss tissue. The *Pseudoamblystegium*-associated
 329 *Nostoc*-like organism was lost shortly after transfer into liquid culture, and was overtaken by an
 330 unidentified, co-cultured non-filamentous cyanobacterium. This may have resulted in a failed

331 PCR amplification of *nifH*, though some 16s data from the *Nostoc*-like organism were recovered
 332 (see below).

333 Whole-community amplification of *nifH* gene

334 Culture-free amplification of cyanobacterial *nifH* from *Hylocomium* moss tissue yielded two size
 335 classes of non-chimeric amplicons, one at the expected size of slightly less than 400 bp, and
 336 another with a size of >500 bp. (Fig. 8). Based on PCR controls from *Nostoc*-free plant tissues
 337 and from our *Anabaena* culture, we presume that the smaller size class represents the intended
 338 target, cyanobacterial *nifH*, and the other represents non-target homologous genes from host
 339 moss tissue (Angel et al., 2018). Both amplicon types were observed in all portions of
 340 *Hylocomium* moss tissue (low, middle, upper), indicating possible cyanobacterial colonization of
 341 moss tissues even when not visible through standard microscopic inspection.

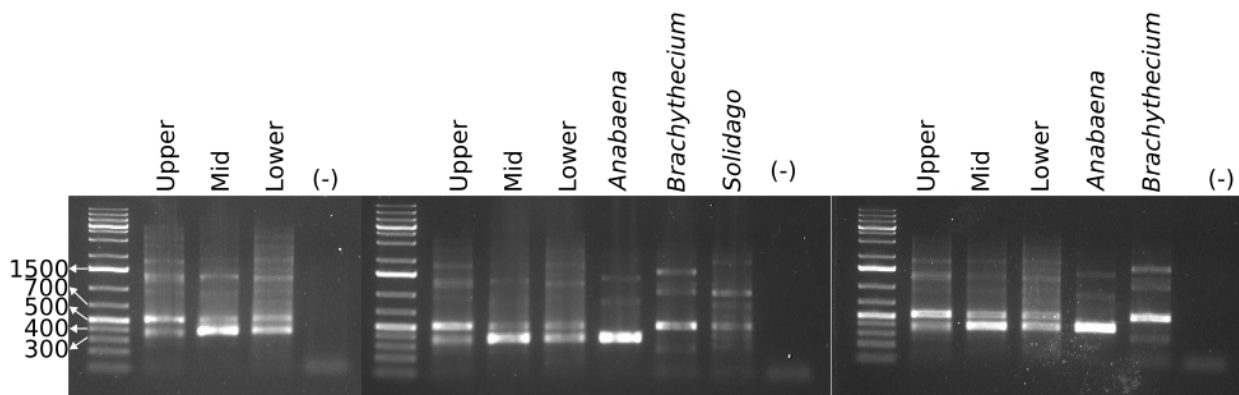


Figure 8. Electrophoresis gel of PCR of *nifH* region, using cyanobacterial specific *nifH* primers on whole community DNA extracted from *Hylocomium splendens*. Bands above 700 bp in size are presumed to be chimeric in origin due to overamplification. Bands in the region of 500 bp are hypothesized to be embryophyta homologs of *nifH*, due to their presence in surface-sterilized, urban *Brachythecium* moss and surface-sterilized *Solidago* leaves, and their absence in axenic *Anabaena* culture. Amplicons of size ~400 are presumed to be the intended the target of PCR reaction, cyanobacterial *nifH*, indicating the possible presence of nitrogen-fixing cyanobacteria in these moss tissues.

342 Phylogenetic placement of *nifH* genes from cultured *Nostoc*-like organisms

343 We detected *nifH* sequences from 3 of the 4 *Nostoc*-like organisms: *Hylocomium*-, *Zygodon*-,
 344 and *Homalia*-associated *Nostoc*-like organisms. Despite differentiated cells resembling
 345 heterocysts (Fig. 6), PCR-amplification of *nifH* from our culture of *Pseudoamblystegium*-
 346 associated *Nostoc*-like organism did not yield any sequence-able product, probably due to
 347 overgrowth by a non-nitrogen fixing cyanobacteria (see above). The topology of our *nifH* tree
 348 recaptured the major clades of Kubota et al. (2023) and Ininbergs et al. (2011) with some minor
 349 differences (Fig. 9, supp. Fig. 2). The most important distinction was the inclusion of the
 350 *Stigonema* *nifH* cluster within *Nostoc* cluster I. In *Zygodon*- and *Homalia*-associated cultures
 351 each appeared to host 3 alternative nitrogenases, each falling into a separate clade on the
 352 Kubota-Ininbergs *nifH* tree. Far fewer high-quality *nifH* sequences were recovered from the
 353 culture of our *Hylocomium*-associated *Nostoc*-like organism, in comparison to our two other
 354 *Nostoc*-like cultures (supp. Fig 1), but these were sufficient to generate a single candidate
 355 consensus sequence for *nifH* in this organism.

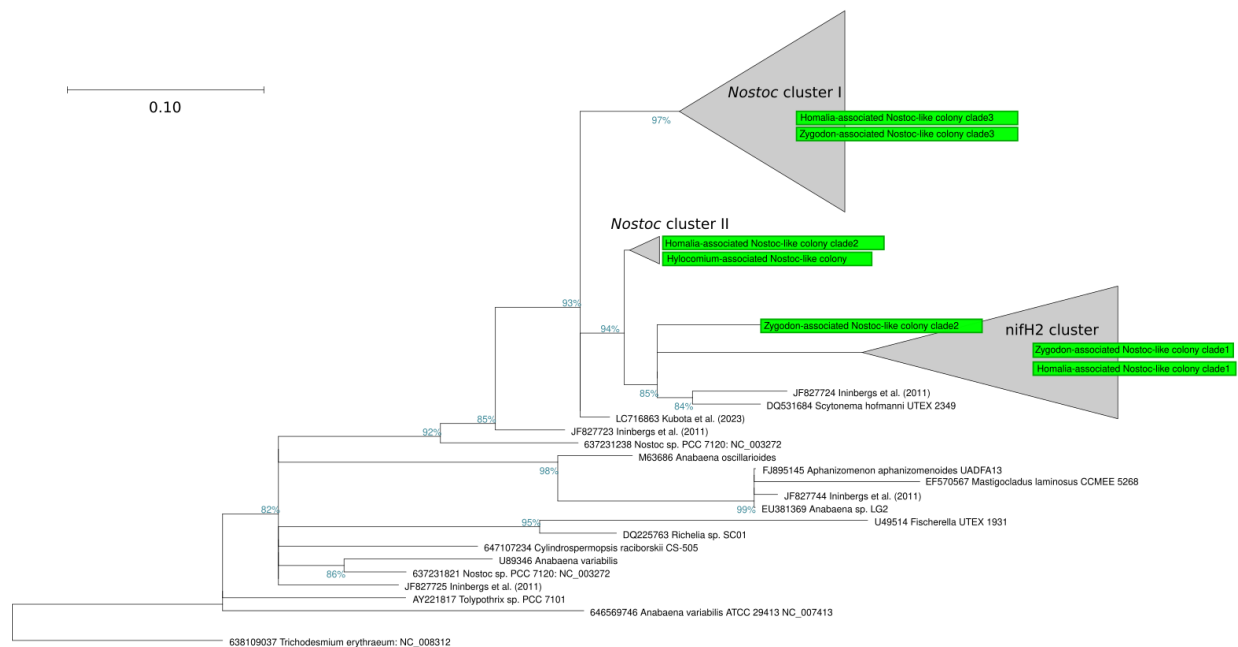


Figure 9. *nifH* gene phylogeny of select members of Nostocaceae. Tree is constructed from de novo alignment of *nifH* sequences detected in this study from *Nostoc*-like organisms (green) and reference sequences from Ininbergs et al. (2011) and Kubota et al. (2023). *Zygodon*- and *Homalia*-associated *Nostoc*-like organisms both hosted multiple distinct copies of *nifH*. Full-size tree with all clades expanded, including placement of "Stigonema Cluster", is available as Supp. Fig. 2.

356 Phylogenetic placement of 16S genes from cultured Nostoc-like organisms

357 Full-length 16S consensus sequences were determined for all 4 successfully cultured *Nostoc*-
 358 like organisms (Fig. 10, green circles). Only one consensus Nostocaceae 16S sequence was
 359 detected for each of the cultures. In the case of *Pseudoamblystegium*, 16S sequence data were
 360 overwhelmingly non-*Nostocaceae* cyanobacterial reads that were not identifiable beyond
 361 phylum-level (Cyanobacteriota), likely due to co-cultured cyanobacteria (see above). However
 362 some high-quality 16S reads remained for identification of the *Nostoc*-like organism originally
 363 observed for this sample.

364 Observed *Nostoc*-like moss symbionts were placed throughout Nostocaceae (Fig. 10, Table 2),
 365 though nearest neighbors were in several cases previously known symbionts: in
 366 *Pseudoamblystegium* and *Hylocomium* mosses, the moss-associated *Nostoc* were closely
 367 related to endosymbionts of macrolichen-symbionts (Miura & Yokota, 2006; Olsson et al., 2012)
 368 and the vascular plant *Gunnera magellanica* (Fernández-Martínez et al., 2013). For *Zygodon*
 369 moss, the cyanobacterium we observed was most closely related to the recently described
 370 genus *Compactonostoc* (Cai et al., 2019). *C. shennongjiensis* was reported as free living on
 371 rocks, another secondary habitat known for *Zygodon* species. For the moss *Homalia*
 372 *trichomanes*, the closest relative was the freshwater-dwelling *Nostoc pruniforme*, as per
 373 Experimental Phycology and Culture Collection of Algae (SAG) strain 62.79, and other free-
 374 living *Nostoc* strains. Closest matches to other 16S sequences from Nostocaceae are shown in
 375 Table 2.

Table 2. Closest sister taxa to observed *Nostoc*-like organisms observed from mosses in this study, based on our de novo 16S tree of Nostocaceae (Fig. 10). When a very close match is available, only one taxa is listed. When a close match is not obvious, the two nearest neighbor taxa are listed.

Moss host	Sister taxa / nearest neighbor
<i>Hylocomium splendens</i>	JN847344 <i>Leptogium</i> -associated <i>Nostoc</i> sp. (macrolichen symbiont)
<i>Homalia trichomanoides</i>	KM019941 <i>Nostoc pruniforme</i> SAG 62.79 (freshwater, free-living)
<i>Pseudoamblystegium</i>	1. KF142669 <i>Gunnera</i> -associated <i>Nostoc</i> symbiont (vascular

<i>subtile</i>	plant symbiont) 2. AB275347.1 Unidentified cyanobacterium Ni1-C1 (unidentified-macrolichen symbiont)
<i>Zygodon dentatus</i>	1. MH598843 <i>Nostoc</i> sp., unpublished, classified as likely <i>Compactonostoc</i> by Cyanoseq 2. KU219709 <i>Compactonostoc shennongjiaensis</i> (Free-living on rocks)

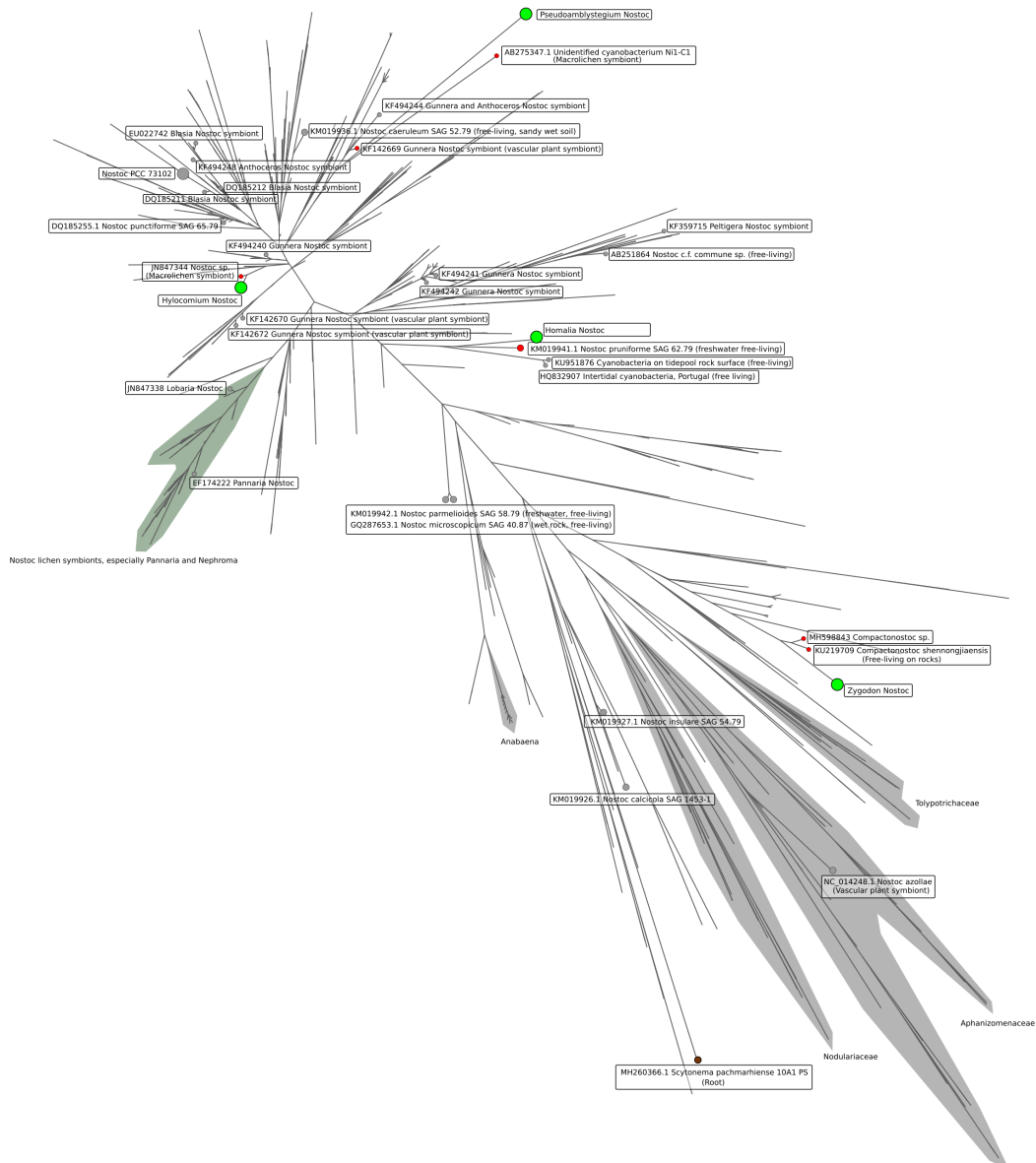


Figure 10. De novo 16S Nostaceae phylogeny constructed from 856 full length 16S sequences from the Cyanoseq database (see methods), selected additional reference sequences, and four 16S consensus sequences from moss-associated *Nostoc*-like organisms from the present study (green circles). Nearest neighbors to moss-associated *Nostoc*-like organisms are shown by smaller red circles, and selected additional reference sequences are labeled by gray circles.

376 Discussion

377 Here we have documented additional instances symbioses of *Nostoc* and *Nostoc*-like
378 organisms with several mosses from the region of middle and upper Franconia, Germany,
379 utilizing a microscopy-, culture-, and barcoding-based approach. Our survey therefore expands
380 the known range of these symbioses further into southeastern Germany. We have also
381 expanded the range of known moss hosts for *Nostoc* to include two non-feather mosses, in the
382 family Orthotricaceae (order Orthotrichales), represented here by *Zygodon* and *Orthotrichum*.
383 Additional new host moss families within *Hypnales* were also documented:
384 *Pseudoamblystegium subtile* (Amblystegiaceae) and *Homalia trichomanes* (Neckeraceae).

385 The finding of *Nostoc* symbiosis in forest-floor *Hylocomium splendens* deeper into central
386 Europe was predicted, since the well-known host moss species *H. splendens* and *Pleurozium*
387 *schreberi* are very common in the region. Nevertheless, the extension of the geographical range
388 of this well-known symbiosis further emphasizes their likely importance to montane forest
389 ecosystems. Given the documented importance of moss-hosted cyanobacterial nitrogen in other
390 systems (discussed above), the contribution of these symbioses to the primary productivity of
391 montane forest ecosystems of our region of Central Europe is also likely to be significant. This
392 argues for the protection of forest-floor moss layers and canopy mosses in the montane forests
393 of the region as part of maintenance of these forest stands.

394 There are obvious disadvantages to a culture-based approach, primarily that presumably many
395 *Nostoc*-like organisms were not successfully cultured. Cultures that were successful were
396 extremely slow-growing, and were in one case not mono-algal. Generally, target *Nostoc*-like
397 organisms were often outcompeted in culture by other photoautotrophs. *Nostoc*-like organisms
398 exist in many other life stages besides colony (syncoebium) stages (Mollenhauer, 1986),
399 including single-cell and single-trichome life stages. *Nostoc* present on mosses in these life

400 stages probably often escape detection by standard light microscopy. Our PCR-tests of moss
401 tissue with cyanobacterial *nifH* genes (Fig. 8) support this, indicating that cyanobacteria are
402 likely present in most aerial tissues of *Hylocomium* even when not detectable with microscopy.
403 As such, the organisms cultured here likely greatly under-represents the diversity of Nostoc-like
404 organisms in mosses in our region.

405 Despite the limitations of our culture- and microscopy-based approach, we documented new
406 moss hosts for Nostoc-like organisms. Particularly of interest are host species *Zygodon*
407 *dentatus* and *Orthotrichum lyellii*, both of *Orthotricaceae* and both of which fall outside of the
408 category of feather mosses (order *Hypnales*). Feather mosses and *Sphagnum* are traditionally
409 understood to be the hosts for moss-Nostoc symbioses, but we show here that such symbioses
410 may be occurring in other distantly-related moss groups.

411 Importantly, these two new host mosses (*Zygodon rupestris* and *Orthotrichum lyellii*) were
412 arboreal mosses. Four out of five moss hosts in which we observed to host Nostoc-like
413 organisms were epiphytic on tree bark (Table 2). We did not conduct systematic sampling for
414 Nostoc-like organisms, so we cannot quantify the relative abundance of the symbiosis in
415 canopy-dwelling mosses in our region vs. forest floor mosses. However, this is supportive of the
416 hypothesized importance of this symbiosis in forest canopies (Lindo & Whiteley, 2011), and we
417 hypothesize that N-fixing cyanobacterial symbionts may provide an important mechanism for
418 allowing moss communities to thrive and compete in the very challenging microclimate of bark.
419 Cyanobacteria in arboreal mosses have been observed previously, i.e. in the coastal temperate
420 rain forest of North America by Lindo and Whiteley (2011) and in a subtropical montane cloud
421 forest in southwestern China for the species *Homaliodendron montagneanum*, *H. scalpellifolium*
422 and *Thuidium cymbifolium* (Fan et al., 2022). Physically smaller and more ephemeral mosses
423 such as *Zygodon* spp. or *Pseudoamblystegium* spp., as well as initial or young stages of larger
424 mosses like *Orthotrichum*, may especially benefit from this symbiosis, as smaller mosses may

not be able to access biologically available nitrogen from atmospheric deposition or stemflow as well as larger mat-forming mosses. If this is true, we would predict symbiosis with cyanobacteria to be especially prevalent on the smaller species as hosts, as we observed for *Zygodon*, *Pseudoamblystegium subtile* and *Homalia*, all of which are comparatively physically small mosses of their families. Whereas the genus *Orthotrichum* is widespread and common in our region, the genus *Zygodon* is rare in central Europe and occurs ephemerally in very scattered mats at sites with a high humidity. *Zygodon* has comparatively small gametophytes and may be at a competitive disadvantage compared to larger, co-occurring species of *Orthotrichum*, and the moss-*Nostoc* symbioses may be especially important for these mosses for continued survival. Additionally, *Zygodon* spp. and *Orthotrichum lyellii* produce large amounts of multicellular gemmae for dispersal. This continual production of gemmae may create additional demand on these mosses for organic nitrogen, and therefore symbioses as we observe here.

We found no clear phylogenetic signal among the symbiotic *Nostoc*-organisms for which the 16S rRNA gene were successfully sequenced - symbionts were placed throughout the *Nostocaceae* family (Fig. 10, Table 2). Nearest matches to other *Nostoc* full length 16S sequences were from various ecological niches, both symbiotic and free-living. Additionally, according to 16S sequence data, our *Nostoc*-like organism isolated from *Zygodon* is most closely related to the recently-described genus *Compactonostoc* (Cai et al., 2019). This genus of *Nostoc*-like organisms has to-date been observed from freshwater free-living states (Cai et al., 2019; Caro-Borrero et al., 2024), but our observation implies that members of *Compactonostoc* may be capable of plant-symbiosis.

Another benefit of our culture-based approach is that we are able to link taxonomic identity (full-length 16S sequences) to a functional gene (*nifH*) within single organisms. This has enabled the observation of possible alternate nitrogenases (3 distinct *nifH* copies) in the genomes of 2 of our moss-associated *Nostoc*-like organisms. Alternative nitrogenases have been previously

450 observed from members of Nostocaceae (Thiel, 1993; Thiel et al., 1995), and are thought to
451 perform supplementary or complementary roles, allowing diazotrophy in different oxygen levels,
452 or the use of alternative metal co-factors (vanadium or iron-only) in place of molybdenum (Zehr
453 et al., 2003). It is worth noting that our only terrestrial moss (*Hylocomium splendens*) did not
454 appear to host Nostoc with alternate nitrogenases, only our arboreal Nostoc were observed to
455 have alternate NifH versions. We speculate that in the forest canopy environment, the
456 availability of molybdenum may be limiting, requiring alternate nitrogenases with other co-
457 factors. Alternatively, the dynamic microclimate of the forest canopy may select for homologs of
458 nitrogenase genes with different environmental optima. Either possibility would enable
459 diazotrophic cyanobacteria to fix nitrogen and continue growth under a wider range of
460 conditions.

461 The possibility that moss-associated cyanobacteria host multiple nitrogenases has implications
462 for the study of moss-associated Nostoc and their ecological services. For example, Warshan et
463 al. (2016) and Arróniz-Crespo (2022) attribute relative importance in terms of biological nitrogen
464 fixation to different cyanobacterial groups in boreal forests, based on differential abundances of
465 sequences falling into nifH clades defined by Ininbergs et al. (2011). However, we show here
466 that great care should be taken when assigning taxonomy using these nifH clusters. Warshan et
467 al. (2016) and Arróniz-Crespo (2022) suggest *Stigonema* as the most important group
468 contributing to fixation of nitrogen in boreal forest systems based on high relative abundance or
469 high levels of expression of nifH genes aligned to the “*Stigonema* cluster” of nifH, sensu
470 Ininbergs et al. (2011). Our results indicate that the Nostoc strain cultured from *Homalia*
471 *trichomanoides*, closely related to *Nostoc pruniforme* SAG 62.79 (Fig. 10), has a version of nifH
472 that falls within this “*Stigonema*” cluster, but also hosts two other nifH genes from other clusters
473 sensu Ininbergs et al. (2011) (Fig. 9 and Supp. Fig. 2).

We also observed the possible endophytic colonization of moss stems by cyanobacteria (Fig. 7). Among mosses, cyanobacterial endophytes are only previously known in *Sphagnum*, which hosts *Nostoc* and other bacteria in specialized structures known as hyalocytes, or dead, hyaline leaf cells (Carrell et al., 2022). Moss leaflet lamina are typically only one cell thick (Roberts et al., 2012), probably precluding abundant endophytic colonization of leaves by microbes that is commonly observed in vascular plants (Bodenhausen et al., 2013; Rodriguez et al., 2009). Stems ('caulids') and rhizoids of mosses are multicellular, and are therefore presumably the site of colonization of much of the extensively documented, diverse non-cyanobacterial endophytes observed in mosses (Tveit et al., 2020; U'Ren et al., 2010). If the observed cells are indeed cyanobacteria occupying the stem tissue of mosses, it points to an abundant symbiosis. We found epiphytic syncoebia relatively rarely in our mosses, so the possibility exists that much of the relatively high rates of nitrogen fixation by cyanobacteria among mosses in our region as noted by Groß et al. (2024) may be achieved by these potential endophytic symbionts instead. Endophytic colonization by *Nostoc* and other cyanobacteria is well documented in other plant- and algal-symbiotic systems, including in hornwort and liverwort hosts (reviewed by Adams and Duggan (2008)). Further experimental work is needed to confirm the cyanobacterial nature of these cells, and if they are capable of nitrogen fixation.

Conclusion

Here we extend the known geographical range of the classical feather moss *Hylocomium/Nostoc* symbiosis further into the region of central Europe, highlighting the importance of moss layers for nitrogen cycling in montane forests of central Europe. We expand the known host range of to include additional non-feather mosses (*Orthotrichum* and *Zygodon*), as well as to other feather mosses not previously observed to host *Nostoc* symbionts (*Pseudoambylestegium*, *Homalium*). We note that multiple paralogs of *nifH* can be found within

498 some *Nostoc*-like organisms, which may denote adaptation to the dynamic environment of the
499 forest canopy, but which also implies that great care be applied when attempting to identify
500 these Nostocaceae symbionts with nifH gene sequence data. We also report a potential site of
501 high cyanobacterial colonization, namely the endophytic compartment of the “stem” (caulid) of
502 mosses. If occurring, this implies an abundant, physically-protected symbiosis but not with
503 *Nostoc*, or at least not in the large-colony stages.

504 Taken together, these results help to bridge the gap between two commonly observed forest
505 moss symbioses: (1) few-to-few, feather-moss/*Nostoc* symbioses, and (2) many-to-many
506 moss/cyanobacterial-community symbioses. From our survey, the former appear to occur in our
507 region but relatively rarely. Based on other reports (cited above), the latter class of broader
508 cyanobacterial symbioses is apparently ubiquitous in mosses where nitrogen is limiting,
509 including in our region. The question now arises as to whether there are particular advantages
510 for mosses to entering into symbiosis with *Nostoc*-like organisms in particular, or whether these
511 feather-moss/*Nostoc* symbioses are simply another case of very common moss-cyanobacterial
512 symbioses. It is also very possible that complex and diverse cyanobacterial communities in
513 mosses have been understudied historically by researchers when highly visible and charismatic
514 *Nostoc* colonies are present. However, in the course of our surveys we noted *Nostoc*-like
515 colonies more often in arboreal habitat, and in locally uncommon mosses that were relatively
516 specialized tree-epiphytic mosses such as *Zygodon*. As such we suggest that host-symbiont
517 specialization could be occurring in the special habitat of the forest canopy. However,
518 quantitative surveys to compare the prevalence of *Nostoc*-like organisms between the among
519 habitats are still mostly lacking, and the unique benefits of such specialized canopy symbioses,
520 if any, are yet to be explored.

521 Declarations

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523 research

524 **Conflicts of interest/Competing interests** - the authors declare no conflicts of interest

525 **Ethics approval** - This article does not contain any studies with human participants or animals
526 performed by any of the authors.

527 **Consent to participate** - not applicable

528 **Consent for publication** - not applicable

529 **Availability of data and material** - code and supporting files are available at the associated
530 github.com repo in the folder "suppData", at the following link:

531 <https://github.com/danchurch/scalingUpByLookingCloser_Feulner_etal/tree/main/suppData>.

532 Demultiplexed, raw Nanopore reads are available at NCBI SRA, from the bioproject

533 PRJNA1401537.

534 Author Contributions

535 MF conceived of the survey topic and design. All authors contributed to field work. MF, US,

536 FJPD, and DCT contributed to microscopic imaging and culturing of microorganisms. JM

537 contributed the hand-drawn illustration of Hylocomium moss and Nostoc (figure 1). MF and US

538 identified moss species. JM, FJPD and DCT conducted laboratory experiments. DCT conducted

539 DNA sequencing and wrote bioinformatic pipelines. MF and DCT wrote the manuscript, and

540 other authors provided editorial input.

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